Method	AppWorld `test_normal`, 168 tasks				PlanBench-XL `test`, 90 queries			
	$\mathbb{E}[U_B]\uparrow$	$\Delta_{\text{evo}}\uparrow$	$\mathbb{E}[Y](\%)\uparrow$	$\Delta_{\text{use}}\uparrow$	$\mathbb{E}[U_B]\uparrow$	$\Delta_{\text{evo}}\uparrow$	$\mathbb{E}[Y](\%)\uparrow$	$\Delta_{\text{use}}\uparrow$
Qwen2.5-7B-Instruct (7.6B, $d = 3584$)								
Frozen Base	0.2583 ± 0.0109	$+0.0000$	34.92	$+0.2042$	0.2633 ± 0.0304	$+0.0000$	35.93	$+0.1167$
AREW / T3	0.2771 ± 0.0127	$+0.0188$	37.70	$+0.2333$	0.2809 ± 0.0218	$+0.0176$	38.52	$+0.1167$
GiGPO	0.3094 ± 0.0299	$+0.0512$	42.46	$+0.2625$	0.3163 ± 0.0395	$+0.0530$	43.70	$+0.1361$
InfoPO	0.3634 ± 0.0289	$+0.1052$	50.40	$+0.3208$	0.3975 ± 0.0447	$+0.1341$	55.56	$+0.2194$
WorldEvolver	0.3636 ± 0.0188	$+0.1053$	50.40	$+0.3694$	0.4441 ± 0.0253	$+0.1807$	62.59	$+0.3208$
PoS+EnvProbe	0.5112 ± 0.0021	$+0.2529$	72.42	$+0.4083$	0.4921 ± 0.0234	$+0.2288$	70.00	$+0.3951$
Step-DPO	0.5583 ± 0.0713	$+0.3000$	79.37	$+0.4958$	0.5356 ± 0.0730	$+0.2723$	75.93	$\mathbf{+0.4142}$
Ablation-Unweighted	0.5842 ± 0.1161	$+0.3260$	**88.49**	$+0.3912$	0.5932 ± 0.0605	$+0.3299$	90.37	$+0.2403$
SPAR (Ours)	$\mathbf{0.6130} \pm 0.1138$	$\mathbf{+0.3547}$	**88.49**	$\mathbf{+0.5033}$	$\mathbf{0.6243} \pm 0.0470$	$\mathbf{+0.3610}$	**91.48**	$+0.3724$
Llama-3.1-8B-Instruct (8.0B, $d = 4096$)								
Frozen Base	0.2610 ± 0.0099	$+0.0000$	35.32	$+0.2139$	0.2683 ± 0.0226	$+0.0000$	36.67	$+0.1167$
AREW / T3	0.2744 ± 0.0141	$+0.0134$	37.30	$+0.2333$	0.2933 ± 0.0326	$+0.0249$	40.37	$+0.1167$
GiGPO	0.3148 ± 0.0338	$+0.0538$	43.25	$+0.2819$	0.3189 ± 0.0350	$+0.0506$	44.07	$+0.1458$
WorldEvolver	0.3688 ± 0.0123	$+0.1078$	51.19	$+0.3694$	0.4441 ± 0.0253	$+0.1757$	62.59	$+0.3208$
InfoPO	0.3838 ± 0.0232	$+0.1227$	53.37	$+0.3500$	0.4127 ± 0.0517	$+0.1444$	57.78	$+0.2486$
PoS+EnvProbe	0.5112 ± 0.0021	$+0.2501$	72.42	$+0.4083$	0.5067 ± 0.0044	$+0.2384$	72.22	$+0.4142$
Step-DPO	0.5782 ± 0.0725	$+0.3171$	82.34	$+0.5153$	0.5567 ± 0.0680	$+0.2883$	79.26	$+0.3746$
Ablation-Unweighted	0.6258 ± 0.0663	$+0.3647$	96.23	$+0.4536$	0.6197 ± 0.0442	$+0.3514$	95.19	$+0.2806$
SPAR (Ours)	$\mathbf{0.6592} \pm 0.0495$	$\mathbf{+0.3982}$	**96.63**	$\mathbf{+0.5453}$	$\mathbf{0.6507} \pm 0.0386$	$\mathbf{+0.3824}$	**95.93**	$\mathbf{+0.3962}$

Table 1: Main performance comparison on held-out test sets across two open-weight LLM backbones (3 seeds, Mean $\pm$ Std). $\mathbb{E}[U_B]$ denotes normalized success-over-time utility, Δ_{evo} denotes self-evolution gain over Frozen Base, $\mathbb{E}[Y]$ is native terminal verification success rate, and Δ_{use} measures development-set masked-feedback utilization.

Diagnostic metric ($p = 0.4$)	Two-sample RLOO	SPAR
Expected objective $J(\theta_0)$	$58/125$ $= 0.4640$	$58/125$ $= 0.4640$
Analytical $\nabla_\theta J(\theta_0)$ bias	0.0000	$\mathbf{0.0000}$
Root-gate variance diff.	Baseline	$\mathbf{-0.0164736}$
Total gradient variance $\sum_i \mathrm{Var}(\widehat{g}_i)$	0.0711904	$\mathbf{0.0395168}$ (-44.5%)
Expected pair cost $\mathbb{E}[C]/B$	0.4800	0.5000
Variance $\times$ cost product	$0.0341714\,B$	$\mathbf{0.0197584\,B}$ (-42.2%)
Cold-start miss rate ($p = 0.02$)	96.04%	$\mathbf{0.00}\%$

Table 2: Exact rational enumeration verification of Theorem 3.1 and Theorem 3.2 on the canonical two-stage partially observable benchmark.